

Deciphering the Platinum Electrolyte Interface in Aqueous Solution

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Computational setup

In this study, we calculated the lattice parameters and bulk modulus for platinum (Pt) bulk.

- Pt slab ($4\times 3\times 4$) with 48 atoms.
- CP2K/QUICKSTEP.
- Goedecker-Teter-Hutter pseudopotentials (GTH-PBE-q10).
- Double Zeta Valence Polarized basis set.
- PBE, RPBE functionals.

Cell size

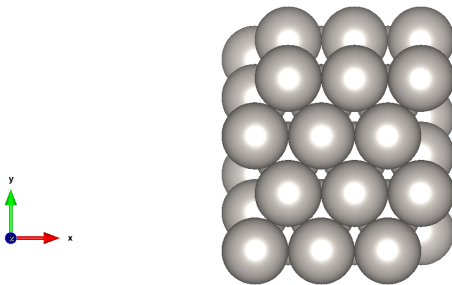


Figure 1: Bulk fcc Pt slab constructed using the Python library *pyiron* for a cell size of 48 atoms with the gamma point.

Cell size

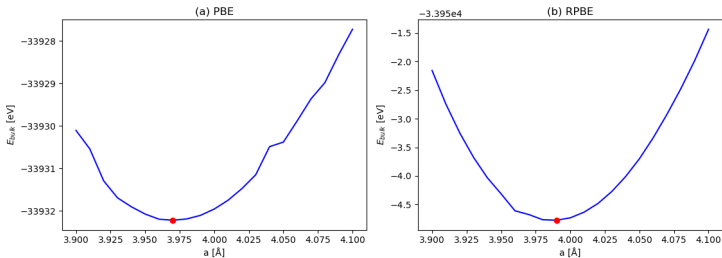


Figure 2: Total energies as a function of lattice parameter a (Å) for Pt (fcc) using different functionals (a) PBE and (b) RPBE with 400 Ry of cutoff and a $8 \times 8 \times 8$ k -mesh.

Cutoff energy

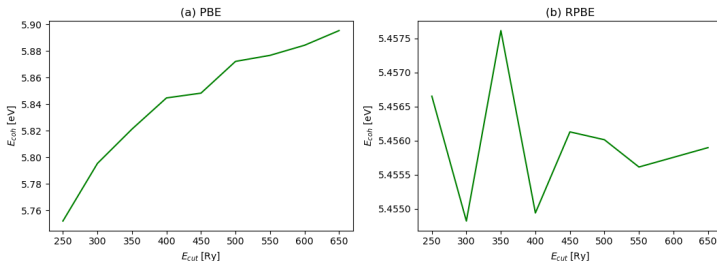


Figure 3: Bulk energies E_{bulk} (eV) as a function of cutoff energy E_{cut} (Ry) for Pt (fcc) using different functionals (a) PBE and (b) RPBE3 with a $8 \times 8 \times 8$ k -mesh.

Bulk modulus

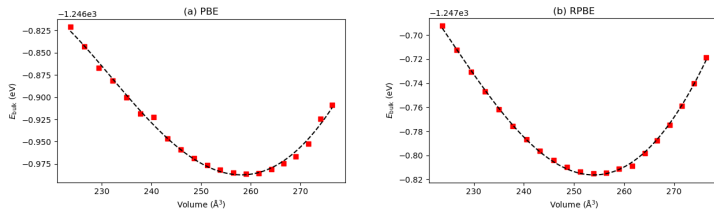


Figure 4: Total energy as a function of volume for Pt (fcc) using different functionals (a) PBE and (b) RPBE, calculated for a $8 \times 8 \times 8$ k -mesh with a cutoff value of 500 Ry.

Lattice parameters

Functional	a (Å)	E_{coh} (eV)	B_0 (GPa)
PBE	3.97 ^a 3.97 ^b	5.84 ^a 5.56 ^b	258 ^a
RPBE	3.99 ^a 3.99 ^b	5.45 ^a 4.90 ^b	254 ^a
Experimental value	3.92 ^c	5.86 ^c	278 ^c

^a This work.

^b Surendralal[1].

^c Kittel[2].

Table 1: Calculated lattice parameters and bulk modulus of Pt (fcc) in a 4x3x4 cell with 48 atoms and different functionals at 400 Ry cutoff.

References



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